Brainstorming Document - Team 6

Ideas (at least 15) :

1. **Data Pals**
    a. **User Population:** Students conducting experiments that requires data collection from a population
    b. **Problem:** Most students stress on meeting a data collection quota for their classes or need an easier way of collecting data.
    c. **Evidence/Assumptions:** Students struggle to find enough participants so a data sharing/data collection platform would help relieve that stress. Then students would be able to participate in other's data collection and exchange help.

2. **Found It!**
    a. **User Population:** College students who lost personal items
    b. **Problem:** Students/Staff that have lost an item on campus that may be hard to financially recover from
    c. **Evidence/Assumptions:** Students struggle with recovering their lost items and campus lost-and-found desks have low recovery rates. Most common items can be dorm keys, student IDs, or tech that can be expensive to recover.

3. **Gigsss**
    a. **User Population:** College students with businesses or gigs
    b. **Problem:** Students with businesses/gigs that they may struggle to market to their community.
    c. **Evidence/Assumptions:** Students with businesses or side gigs struggle to market their services to the community or commitments may make it difficult for them to do so.

4. **Laundry Helper**
    a. **User Population:** College students living on-campus
    b. **Problem:** Laundry machines are always being used in the school dormitory which makes it challenging for residents to find available washers and dryers.
    c. **Evidence/Assumption:** By allowing students to see what washers and dryers are available and the time remaining on the machines, students are able to save time and energy from making multiple trips around their dorm building. This helps mitigate the laundry traffic.

5. **Elevator Wait Times**
    a. **User Population:** Accessibility students, and students who don't want to use the stairs in general.
    b. **Problem:** Students are having to wait in line constantly to use the elevator in busy buildings during peak class times, but it's annoying to not know whether or not it is worth it.

    c. **Evidence/Assumption:** Helps save students time, and can be helpful for students with disabilities to know when to head for the elevator.

6. **Free Food!**
    a. **User Population:** Students on campus.
    b. **Problem:** Everyone loves free food, but free food popups are so random on campus that many students end up missing them.
    c. **Evidence/Assumption:** This idea could help save students money and give them opportunities to try different cuisines.

7. **Handicapped Button Down**
    a. **User Population:** Accessibility students.
    b. **Problem:** Sometimes the handicapped button for doors is down, and you never know when.
    c. **Evidence/Assumption:** This project will help handicapped students, or even students with both hands preoccupied know where they can enter buildings or if they need to take an alternate route.

8. **Which Study Room? → Studii**
    a. **User Population:** Students.
    b. **Problem:** During peak midterms and finals season, the library can get very packed. Students sometimes are overwhelmed with trying to decide where there is space for them to study.
    c. **Evidence/Assumption:** This could help students find a study space faster and more calmly.
9. **Repairing Schools**
    a. **User Population:** All students around campus.
    b. **Problem:** It's common to find broken or damaged facilities or items around campus, but it's unclear where or who to report them to.
    c. **Evidence/Assumption:** Providing an anonymous way to report issues will encourage students to speak up about things that need to be fixed, ultimately improving campus life for everyone.
10. **Item Exchange**
    a. **User Population:** Students who need an item but can't or don't want to purchase it.
    b. **Problem:** Students often want to borrow items for short-term use, but they don't have anyone in their personal network to borrow from.
    c. **Evidence/Assumption:** Facilitating an item exchange saves money and reduces environmental waste caused by purchasing single-use items.
11. **Silent Help**
    a. **User Population:** Shy students in classroom settings.
    b. **Problem:** Students are often afraid or hesitant to speak up in class when they are confused or need clarification.

c. **Evidence/Assumption:** Giving students a button to signal professors or TAs silently will prompt instructors to slow down or re-explain topics, allowing all students to participate comfortably.

12. Chore Exchange

a. **User Population:** Students who dislike certain chores and enjoy others.
b. **Problem:** Students often struggle with chores they dislike, but lack an easy way to swap tasks with peers who prefer those specific chores.
c. **Evidence/Assumption:** Allowing students to volunteer, trade, or get paid for doing chores for one another makes routine tasks easier to manage and fosters a helpful community on campus.

13. Barnumbia Events

a. **User Population:** First-year students and students who are unaware of campus events.
b. **Problem:** Campus events can feel exclusive or overwhelming to navigate, making it hard for students to find activities that fit their interests and schedules.
c. **Evidence/Assumption:** Centralizing and tracking event details makes campus activities more accessible, ultimately improving the social life for students across all class years.

14. What's Available?

a. **User Population:** Campus commuters, late-night studiers, and budget-conscious students.
b. **Problem:** Finding open dining halls, late-night food options, or available campus food pantry items during non-peak hours is inconvenient and often leads to wasted trips across campus.
c. **Evidence/Assumption:** Providing real-time updates on food availability, operating hours, and surplus campus meal drops helps students stay nourished, saves time, and reduces food insecurity.

15. **Happy Hour Finder**

a. **Target user population:** Older students looking for a good time with friends nearby, in need of a place with a happy hour near campus/in Manhattan.
b. **Problem:** Students who are new to the city may not know "the spots" near Columbia campus. Students may not know where to go or overpay for drinks. Information is scattered and hard to find.
c. **Evidence/Assumption:** Centralizing deals and good locations will make planning social outings more affordable and easy to find/do. This will also support local businesses around Columbia.

16. **Tech/Furniture/Appliances green marketplace**

a. **Target user population:** Columbia students looking to recycle and reuse other students' belongings and a place for students to donate or sell all sorts of things.
b. **Problem:** Students often need appliances and/or other items when moving in or when something breaks; on the flip side, students may no

longer need certain items. Some donation places exist, but mid-year, this is a problem, and there is no direct way to bridge the gap.

   c. **Evidence/Assumption:** More recycling and less waste is good; it also creates convenience for students at Columbia.

17. **Party planning app with proximity awareness and optional public settings (under the impression that it's only for CU students)**
    a. **Target user population:** Columbia students organizing and planning private or public social events.
    b. **Problem:** Students often rely on word of mouth for certain parties and events. Although some solutions already exist, accessibility remains a problem. Having something location-based and with the addition of publicly available (larger events) will create more exposure and make it easier for students to be involved.
    c. **Evidence/Assumption:** A Columbia-only platform with proximity awareness would make it easier for gatherings to gain traction and give students more opportunities to connect with other students.
18. **Dodge gym buddy application**
    a. **Target user population:** Columbia students who want to have motivation to stay consistent with their gym schedule and make new friends.
    b. **Problem:** Students may feel unmotivated to go to Dodge for a number of reasons–whether it is not knowing their way around it or maybe they need someone to motivate them and hold them accountable.
    c. **Evidence/Assumption:** Buddying students with similar goals is a great way to foster connections/friendships and promote wellness.

****Everyone NOTE that each of the things we have to get done should be done as separate files, and we collect them all and then turn them in as a bundle.****

1. Contextual Inquiry Notes - Olivia will do, and share with whoever needs to do an affinity diagram.
2. Affinity Diagram – Ismael
3. Journey Map – Ismael
4. Storyboards - Nicole
5. Comparative Analysis - Emily finished

CHECKPOINTS FOR VIDEO:

1. Project recap

    a. What is your project called?

    b. What is your project about?

    c. Where was your project at the beginning of this checkpoint?

2. Activities and progress: - Olivia

    a. What did you do during this checkpoint?

    -

    b. How has the contextual inquiry shaped your team's thinking about the Project?
    (Has it shaped your thinking about what the problem statement or target user population actually are?)

    - It has helped us find where the pain points are of the current UI. How students are getting frustrated and with what.
    - Frustrations lead to a more hectic experience, which is only exacerbated by the stress of school/midterms/finals.

    c. Which design solution concept (storyboard) do you feel has the most

promise, and why?

- Campus Map, good visuals, different from the list/grid that exists.
- Possibly more intuitive

3. Risks

a. What is your plan for testing the concept represented by your storyboard? Please be specific.
(If it is a smoke-and-mirrors prototype, describe what that prototype might be. Please be specific and refer to the “Concept testing plan” grading rubric section to see what we are looking for. We are asking this now to give teams feedback early, before Checkpoint 3 starts in earnest.)

b. What are your project’s biggest risks as of now, and how do you plan to mitigate them in the next two weeks?
(These can be any type of risk such as design, technical, or scheduling / project management related.)

4. Questions for your mentor

a. Any specific questions or advice you would like?
How can we keep track of crowd-density?

Contextual Inquiry - Observations, Notes, Personal Accounts

NOTE: Below is the collection and organization of my notes from interviewing two users in our target population as they used the current study room booking system, in the middle of midterm season (many rooms were booked, this was their first time seeing this).

After observing and interviewing the two participants from our target population, I have discovered some notable differences in the way the team was at first thinking of our problem. As we first believed in the problem being perhaps an issue of there not being an existing, intuitive UI that allows students to know when their favorite study room (or any study room, especially during midterms/finals season) is free at that exact moment, it seems that there are actually some underlying issues of misconception and misunderstanding with the current UI that can help inform our design.

For example, the two different participants showcased opposite reactions to the UI. Participant 1 said that she did not understand the UI at all, growing increasingly frustrated as they tried to book a study room within the next few days (which is in the middle of midterm season, hence there were not that many options). They at one point stated:

> The website is not intuitive because there's like two tabs for each study room. There's like the ones on the main floor (in reference to the Uris study rooms) and then there's like a second floor of study rooms and its two separate tabs. You have to check both which is annoying and then it's just like such a process and like that, I don't know just having to pick the timeslots and I can never read it properly.

This goes to prove that our UI will be an important factor in ensuring that our target population is not deterred by any avoidable frustrations that might hinder their use of our prototype, and to find their room quickly. Another element to note is that there is more misinformation regarding knowledge of room availability, something that we strive to solve with our project. Participant 1 believed so, by stating:

> I literally don't know what study rooms are available except for the ones on Uris. I don't know if there are other rooms. I'm sure there are. I wouldn't know. I don't know if Columbia has a website for that. I've never seen it.

Participant 1 seemed to be a big believer in some centralized form of communication for others to be notified about specific rooms being booked, and they said later while frantically attempting to find any room that was free, “people are always walking in on me and they're like, ‘no no I booked it.’ I don't think that they did, so I don't enjoy that process.”

With Participant 1, the evidence is clear. The current booking system is not straightforward, it frustrates current users, and even so, there are still miscommunications with which rooms are booked and the demand for there be notifications given to students about study-room-availability around campus.

Now, it is noteworthy to study the reactions that Participant 2 gave after interacting with the current study room booking system. This participant had an entirely different interaction with the system, noting that they thought the process was “incredibly easy.” During my brief 10-minute interview, I found this stark contrast interesting and asked their frequency in using study rooms. They responded saying they book rooms almost every week. However, they did say that they often “get mixed up on the room [they] are booking.”

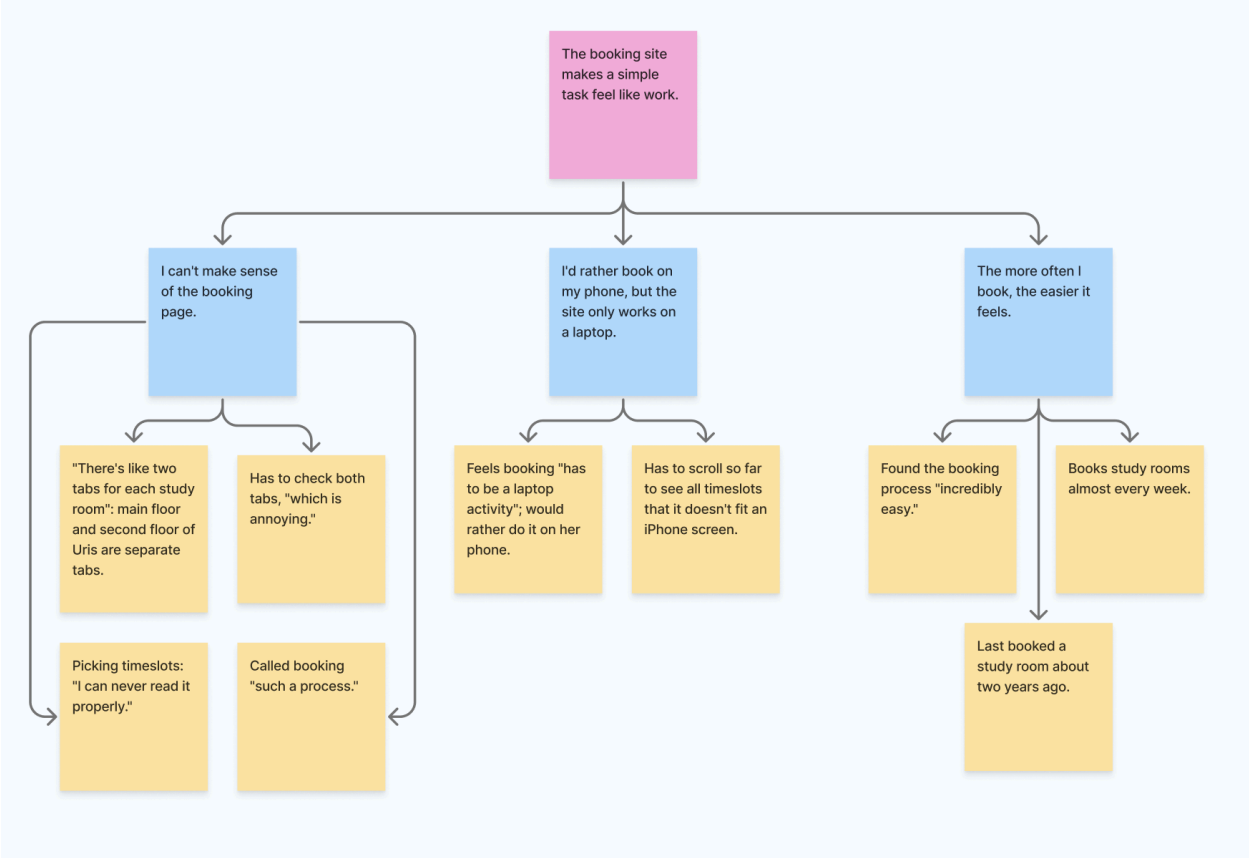
This statement caught my attention, as I asked whether or not they would like to have some sort of system that would give notifications about the availability of study rooms, keeping the study room’s status fresh in their mind, and Participant 2 responded with positive reviews stating, “that would be something I would be interested in. Especially since I use the rooms so frequently, I have grown to like some rooms over others. It would be cool to know when my favorite room in Uris is free when I’ve just seen my favorite room in Butler is booked.”

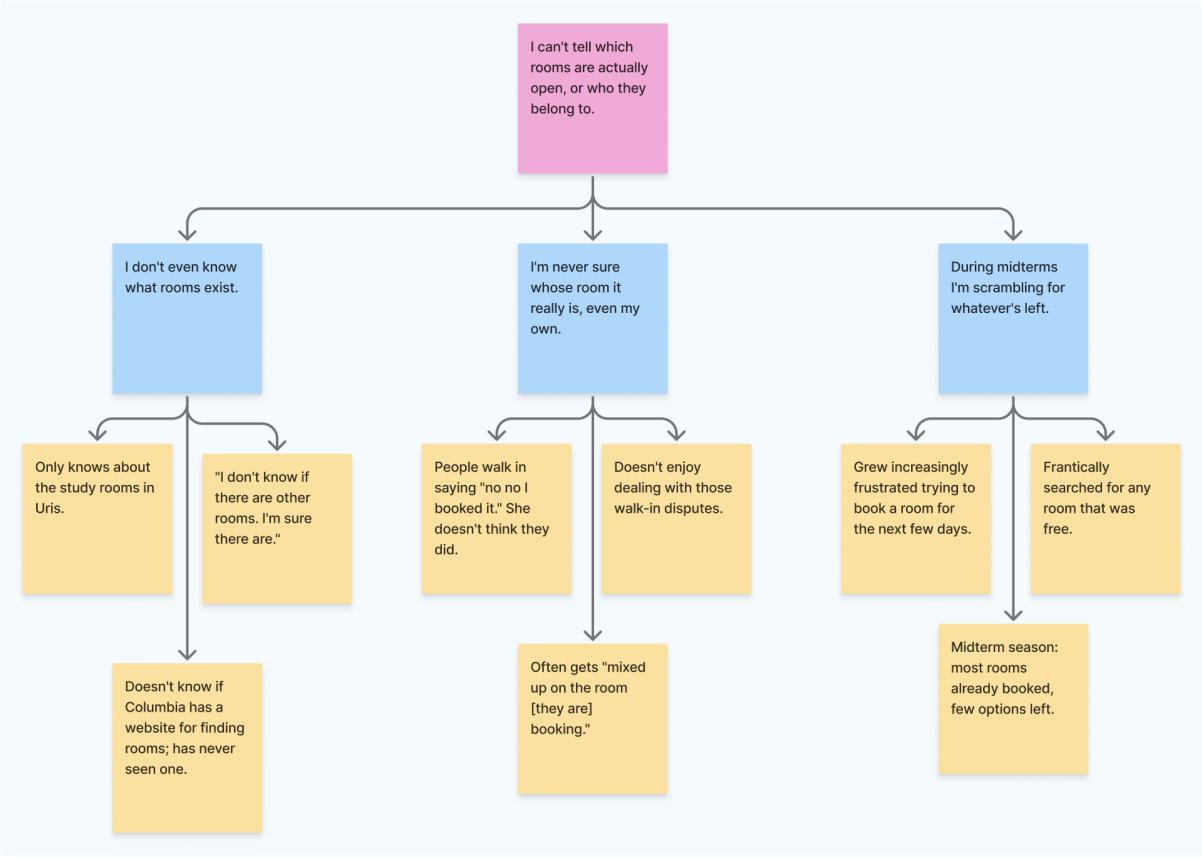
<u>BONUS OBSERVATION BELOW!</u>

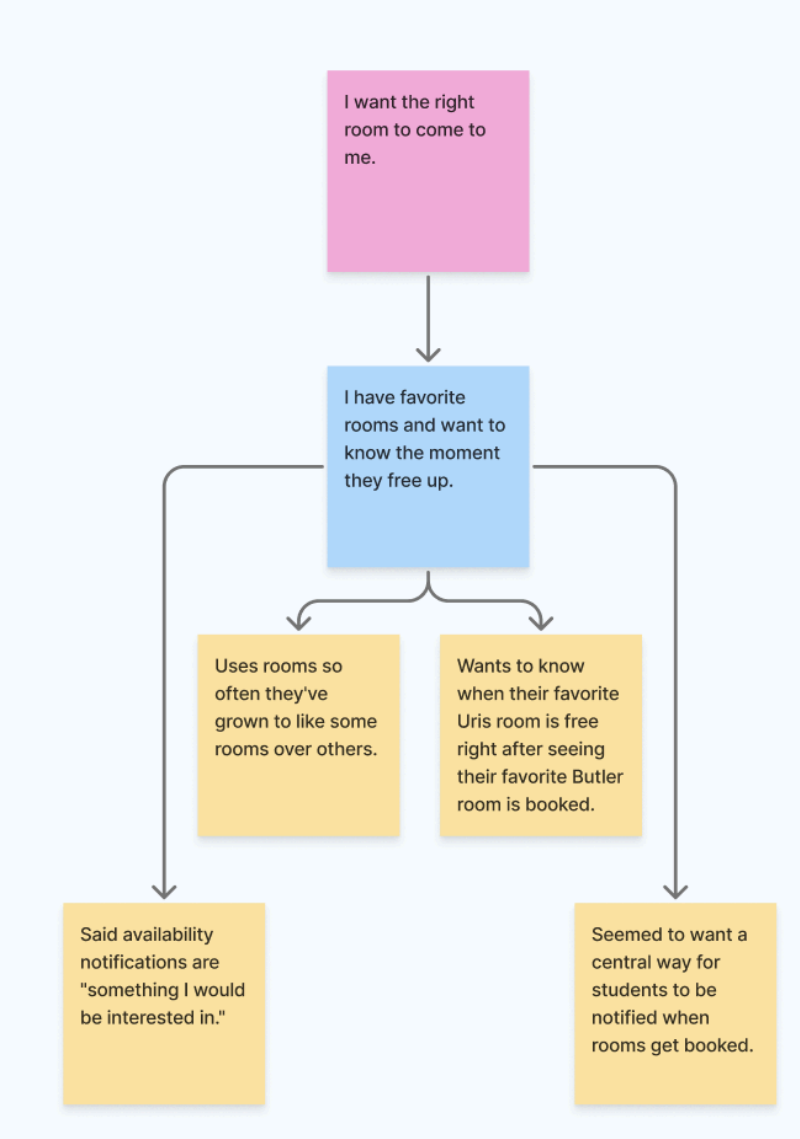
Quote from the interview w/ Participant 1 that might be worthy of a read:

- “Oh my other thing is I feel like booking a year a study room has to be a laptop activity and I'd rather be able to do it on my phone but at least in my experience whenever the last time I did, it was like two years ago like do you have to scroll so far to see all the timeslots that it doesn't work with the aspect ratio of an iPhone you have to do it on the computer and that's annoying” - Participant 1
    - This could mean that perhaps the project should be more geared towards phone users than laptops, to ensure easy/more realistic access.

Affinity Diagram





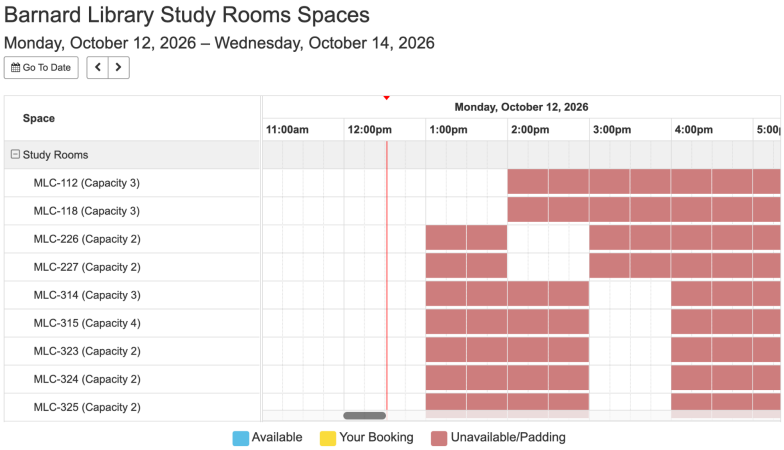

I want the right room to come to me.
I have favorite rooms and want to know the moment they free up.
Uses rooms so often they've grown to like some rooms over others.
Wants to know when their favorite Uris room is free right after seeing their favorite Butler room is booked.
Said availability notifications are "something I would be interested in."
Seemed to want a central way for students to be notified when rooms get booked.

Journey Map

https://docs.google.com/spreadsheets/d/1Ys8sCUjoKmFXGxnhS16WZCl9xYSvO0yg0uwBxHp_sNA/edit?usp=sharing (Log in with CU email)

Solution 1: Libcal

Our university currently uses LibCal as the administrative booking system where students can reserve study rooms at a given time slot (up to 290 minutes per day). To reserve this space, students need to go through a multi-step authentication to have that space under their name for that time slot.

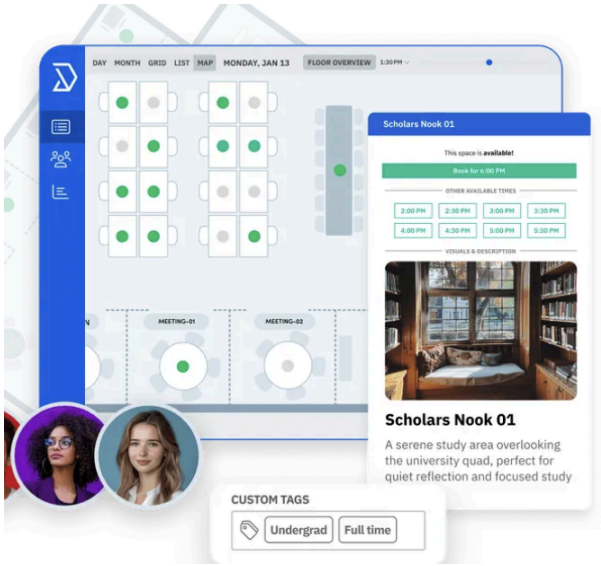


In the screenshot above, the system does well with guaranteeing students the space once they reserve the spot. It also does well with giving a time cap on the study room, so everyone has the chance to get the space. Librarians or staff are able to keep track of the space usage. However, this system is only useful for students who have study times reserved in advance, so it doesn't accommodate students who need to sit down immediately. This would force those students to go through that multi-step authentication process while also trying to find an available space that hasn't already been taken.

As seen in our contextual inquiry notes, students are most frustrated with the interface of the reservation system and we can see that this design forces students to scroll to find their desired study time. For Columbia's LibCal system, students can only see other libraries by changing it through a tab. This design flaw caters to students who are booking on laptops which ruins spontaneity since it would be harder to find a time on phones. It's also inefficient since students have to scroll through all of the booked times just to find available times.

Solution 2: Skedda

Skedda is another solution that is similar to LibCal but has more use cases where students can reserve more than just study rooms such as desks, meeting rooms, labs, etc. In the screenshot below, the system's design does well with providing the user with an interactive map where they can view available seats. Additionally, the solution uses WiFi-based presence and QR code check in to confirm if the space is actually being used or not.



While Skedda seems to provide the most optimal user interface, its administrative overhead is extremely restricting for students who want to find a spot quickly. The bookings rely on users to go through their account to create these reservations which may lead to frustrating experiences. The design also seems to accommodate a laptop rather than phone, so similar to LibCal in which the design doesn't consider usability on the phone.

Our proposed solution, Studii, is more effective as it gives students two important features: spontaneity and efficiency. Unlike Skedda and Libcal, Studii focuses on giving students a more casual experience where they can conveniently find spaces without having to go through an extensive reservation journey. Students will be able to book their study room times without having to scroll through unavailable times, and immediately know what libraries have available study rooms. For check-ins, instead of having students use a QR code check-ins like Skedda, Studii would simply notify the student of their reservation time and prompt the user to confirm their occupancy. Simply tapping a confirmation button would make the experience smoother so students can focus on their study session. Studii bridges what Skedda has implemented with their user interface and LibCal's administrative overhead.

Storyboards

